Algebra II with Trigonometry Chapter 6.5

Olympiad Problems (GPT-5.4)

Saved tutoring response with MathJax rendering. Original PDF file: 4533652_alg-2trig-h-chp-6-5-note.pdf

Problems

1. Evaluate exactly:

$$\arcsin\left(\frac{3}{5}\right) + \arccos\left(\frac{12}{13}\right) + \arctan\left(\frac{1}{3}\right).$$

2. Compute the exact value of

$$\tan\left(2\arcsin\left(\tfrac{3}{5}\right) - \arccos\left(\tfrac{5}{7}\right)\right).$$

3. Find the exact value of

$$\arcsin\left(\frac{4}{5}\right) - \arctan\left(\frac{4}{3}\right).$$

4. Simplify to a single constant:

$$\arccos\left(\frac{1}{3}\right) + \arccos\left(-\frac{1}{3}\right) + \arcsin\left(\frac{5}{13}\right) + \arcsin\left(\frac{12}{13}\right).$$

5. Let

$$x = \arctan 2 + \arctan 3 + \arctan 4.$$

Determine the exact value of $\tan x$.

6. Find the exact value of

$$\cos\left(\arcsin\left(\frac{7}{25}\right) + 2\arctan\left(\frac{1}{3}\right)\right).$$